CHARGE-IT: Investment Cost Estimation Tool for Overnight Charging Infrastructure of Heavy-Duty Vehicles in Europe

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Keywords: Heavy-duty vehicles · Depot charging · Investment cost modeling · Charging infrastructure · Grid connection · Techno-economic analysis · Decision-support tool

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Abstract

The deployment of overnight charging infrastructure for battery-electric heavy-duty vehicles (HDVs) is critical for decarbonizing European freight transport, yet decision-makers lack transparent, component-level cost estimation tools. This paper introduces an open-source, web-based investment cost calculator for overnight depot and public parking charging stations. The tool takes site-specific electrical parameters as input (number and power of chargers, grid connection type, voltage levels, cable distances, and site conditions) and automatically determines the project type (one of four categories based on grid connection requirements). It then sizes every electrical component from the grid connection point to the charging plug and computes itemized costs using real-world pricing data. The tool supports three analysis functions: cost breakdown, single-variable sensitivity, and two-parameter sensitivity analysis, enabling users to explore how different design choices affect total investment cost. Applied to nine case studies across Germany, Italy, and the United Kingdom, the tool produces investment costs ranging from €20,000 to €80,000 per charging plug, depending on grid connection type, charger power level, and deployment scale.

Code Metadata

Field	Value
Current code version	v1.0.0
Permanent link to repository	https://github.com/davide-ferraro/ev-charging-cost-estimator
Legal Code License	MIT
Code versioning system	git
Languages, tools and services	Python, Streamlit, Plotly, Pandas, NumPy, OpenPyXL
Operating environments and dependencies	Python ≥ 3.10
Link to developer documentation	*To be added*
Support email	davideferraro275@gmail.com

1 Motivation and Significance

Battery-electric trucks are entering the European market at a pace that the charging infrastructure has not kept up with. Heavy-duty vehicles (HDVs) account for over a quarter of EU road transport greenhouse gas emissions and roughly 6% of total EU emissions, despite being a small share of the fleet [1]. The EU's "Fit for 55" package now mandates CO_2 reductions of 45% by 2030, 65% by 2035, and 90% by 2040 for new HDVs [1]. Since electric trucks produce no tailpipe CO_2, they count as zero-emission under these rules, which is a direct incentive for manufacturers and fleet operators to go electric [2]. Battery costs have fallen enough that electric trucks now work across most freight segments [3,4], and McKinsey expects most e-truck operators to prefer overnight depot charging for its predictable schedules and cheap off-peak electricity [5].

The problem is on the infrastructure side. A single long-haul truck battery can hold 300–800 kWh, and a depot charging multiple trucks at once may need several megawatts of grid capacity [3,4,6]. The Alternative Fuels Infrastructure Regulation (AFIR) requires EU Member States to install at least 3.6 MW of truck charging capacity every 60 km along the core road network by 2030 [7]. Europe had roughly 10,000 public truck charging points in 2023; that number needs to reach 300,000 by the end of the decade, at an estimated cost of €7 billion [8]. Most existing depots cannot supply this kind of power, and getting a grid connection upgraded is slow, expensive, and tangled in permitting, sometimes taking years [9].

These infrastructure challenges are compounded by a lack of reliable cost estimation tools. Charging infrastructure for trucks requires high-power chargers, heavy electrical equipment, and site-specific civil works. Fleet managers and investors rarely have clear numbers for how costs change across different configurations [10]. For instance, how much more a depot serving fifty trucks costs compared to one serving five, or what the price difference is between a site that already has a medium-voltage (MV) connection and one that needs a new one built from scratch. Borlaug et al. [10] found that smaller U.S. depots can often avoid costly grid upgrades, but larger sites typically need high-power connections and new substations. The ICCT notes that transformer sizing, permitting rules, and site characteristics create wide cost variation even within a single country [6]. Burton et al. [11] showed that utilities can cut grid-related costs by sizing components around actual truck usage patterns rather than peak demand. Managed charging at the depot works the same way from the operator's side: staggering the sessions lowers the coincident peak, and with it the grid connection that has to be built [26]. Still, the literature mostly reports cost in broad categories ("equipment," "labor," "grid fees") without breaking it down to individual components like cable cross-sections or switchgear ratings [4,8]. Most studies also focus on North America [10,12], where electricity pricing, grid regulations, and equipment standards differ from Europe. Lead time and permitting data is scarce as well [9,13].

Several software tools and models address parts of the EV charging cost problem, but they fall into three groups that each leave the target case uncovered. The first group computes the levelized cost of charging (LCOC), a per-kilowatt-hour operating metric rather than an upfront capital cost. Borlaug et al. [10] built the reference LCOC framework for the United States and released it as the open-source lcoc-ldevs tool, but it targets light-duty vehicles and folds equipment and installation into an amortized input rather than producing a component breakdown. Lanz et al. [23] applied a similar LCOC approach across 30 European countries, yet their model covers passenger transport only and deliberately leaves grid connection costs

out. The Atlas Public Policy charging cost calculator [24] works the same way, estimating the electricity cost of a given charging pattern rather than the hardware and connection investment. The second group is the set of planning and simulation tools in the U.S. Department of Energy EVI-X suite [22]. These project charging demand, simulate site energy use, or run financial scenarios, and they are explicit that capital cost estimation is not their job: the financial tool EVI-FAST asks the user to supply equipment and installation costs from vendor quotes rather than deriving them. Open-source interactive tools in the same family take aim at charging operation or grid impacts rather than capital cost: datafev [28] is a Python framework for charging management and scheduling algorithms, and EV-EcoSim [29] is a grid-aware co-simulation and optimization platform that does quantify cost, but as the levelized cost of grid and distributed-energy-resource components in a United States, light-duty context rather than a component-level capital breakdown of the station from the grid connection to the plug. The third group is the static, component-level cost literature. The AFDC report on non-residential charging equipment [12] and the ICCT metropolitan-area analysis [25] both split installation cost into hardware, labor, permitting, and grid connection, but they are published as fixed tables and figures rather than interactive software, and both describe U.S. passenger charging with data that is now roughly a decade old.

Work specific to heavy-duty depot charging is more recent and still leaves the niche open. Borlaug et al. [26] examined how depot charging stresses the electricity distribution system and released a load-profile generator, but their focus is grid upgrade cost and load shapes rather than a station-level investment breakdown. Wang et al. [27] come closest in intent, building a component-level bottom-up cost model for medium- and heavy-duty depot charging in California, yet they publish it as a methodology and dataset rather than a reusable interactive tool, and the costs are specific to the United States. FleetRL [14] is a reinforcement-learning environment for scheduling fleet charging and minimizing electricity cost; it models charging operation, not the physical infrastructure or its capital cost, so it complements rather than overlaps with the present tool. Across all of these, no open-source, interactive tool estimates component-level investment cost from the medium-voltage grid connection to the charging plug for European heavy-duty overnight charging, which is the gap this software fills.

This paper fills that gap with the first open-source, web-based tool for estimating overnight charging infrastructure investment costs for battery-electric HDVs in Europe. The tool models 15 individual cost components from the grid connection to the plug, using real equipment pricing data in 2024 euros from catalogs, suppliers, and published studies. The hardware the tool prices is not tied to the time of day: a station sized for overnight charging can serve daytime or opportunity charging as well. The overnight label refers to the design case. Trucks parked for the daily rest period of at least nine hours required by Regulation (EC) 561/2006 [30] can be recharged at 20 to 100 kW per plug, so the tool covers the 10 to 350 kW range and leaves megawatt-class en-route charging outside its scope. A simultaneity factor lets the user size the grid connection for managed charging rather than for every plug at full power. It automatically classifies projects into four types based on grid connection requirements: Type 1 where the existing connection already has sufficient capacity, Type 2 where a small LV upgrade covers the gap, Type 3 where a new MV connection is needed alongside an existing on-site transformer, and Type 4 where a full MV buildout is required from scratch. The type determines which cost components are activated. The tool supports cost breakdown, single-variable, and two-parameter sensitivity analyses, and runs as a Streamlit web application that requires no

installation and produces interactive charts and downloadable reports. It was developed at KTH Royal Institute of Technology with Scania AB, with cost assumptions informed by interviews with nine industry professionals across Europe.

2 Software Description

The tool is a Python web application built on Streamlit. Users configure a charging station, run one or more analyses, and get itemized cost results they can export.

2.1 Software Architecture

The tool follows a three-layer architecture. The **presentation layer** (`app.py`, 995 lines) collects the 16 user inputs through a Streamlit widget form. It re-renders the interface on every change to enable or disable fields that depend on the current configuration, and it displays results as interactive Plotly charts with Excel and PDF export. The **orchestration layer** (`utils/calls.py`, 325 lines) validates the inputs and runs `medium_requirement()` and `hard_requirement()` to check whether the existing grid connection can handle the planned load. It then calls `case_definition()` to assign one of four project types and `compute_all_costs()` to dispatch to every relevant cost function. The **computational layer** (`utils/cost.py`, 490 lines) contains the 15 cost functions. Each function takes one or more physical quantities and returns a cost in 2024 euros, using lookup tables or fitted curves from industrial catalogs and published studies. The layer also provides a few shared helper routines for table lookup and unit conversion.

The data flow follows the three layers, as shown in Figure 1. In the presentation layer, the user provides the 16 input parameters. The orchestration layer then classifies the project type by comparing the planned apparent load against the existing grid connection, returning one of four types. This project type acts as a control signal that activates or deactivates specific sizing calculations and cost functions, so components that do not apply to the configuration are switched off. In the computational layer, each active component is sized from the inputs into the electrical quantities the cost functions require (current I [A], apparent power S [kVA], total power P [kW], and voltage V [kV]), and each of the 15 cost functions maps these to a cost in euros. Every active cost function returns one investment-cost line item, while inactive components return zero, and together they yield the final itemized cost breakdown.

Because Streamlit re-executes the script on every widget change, the project type classification and the field-enable logic run automatically on every interaction, without any explicit callback wiring. The electrical sizing and cost calculations, by contrast, run only when the user clicks Run Analysis.

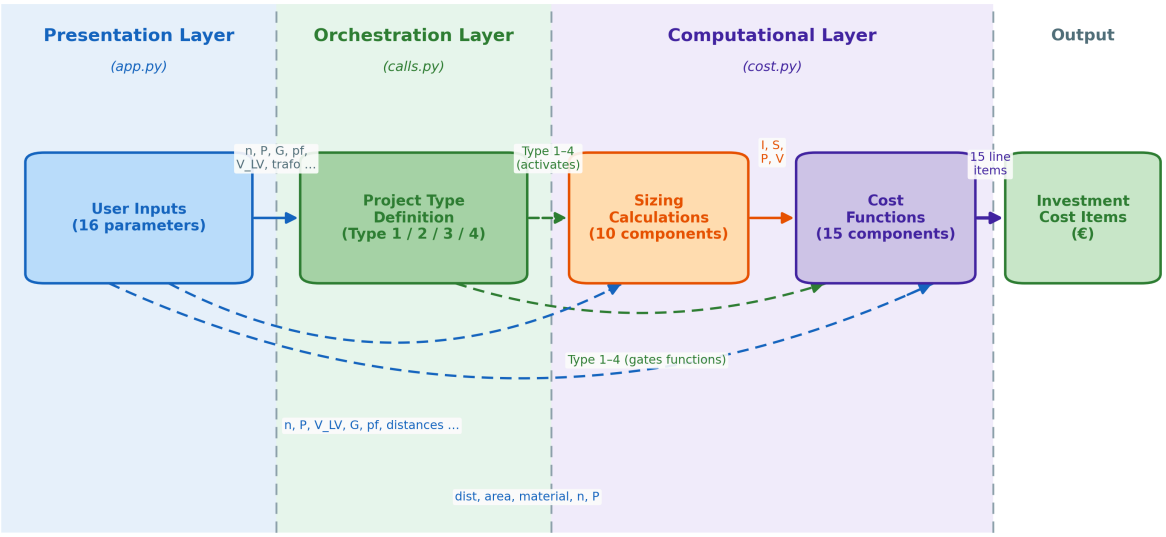


Figure 1: Data pipeline overview. The three architectural layers (`app.py`, `calls.py`, `cost.py`) are shown as bands, carrying the user inputs through the project type classification and the electrical sizing and cost functions to the itemized investment cost.

2.2 Software Functionalities

Input Configuration. The tool allows the user to configure a charging station through 14 numerical and 3 categorical inputs. The charger parameters are the number of chargers n, the power per charger P [kW], the load power factor pf, and the simultaneity factor f, the share of the installed charger power drawn from the grid at the same time. These drive the planned apparent load calculation and appear in nearly every cost function. The grid and connection parameters cover the existing grid connection capacity G [kVA], the LV and MV voltage levels, the available transformer capacity, the maximum LV connection power, whether a transformer is already present on site, and the transformer safety margin. Together these determine the project type and size the MV and LV electrical equipment. The distance and site parameters capture the cable runs from the rectifier to the chargers, from the premises to the nearest MV/LV transformer, and to the nearest MV access point. They also include the land area, pavement material, and terrain type, which determine the cable and site preparation costs.

The planned apparent load is computed as $S = f \cdot n \cdot P/\ \mathrm{pf}$ [kVA] and compared against the existing grid connection capacity G. With $f = 1$, the default, every plug delivers its full power at once, which is the worst case for the grid connection. Lower values represent managed charging, where a site controller caps the coincident load. The grid connection, transformer, switchgear, and distribution cables are then sized on the reduced load, while the chargers, rectifiers, and cabling to the plugs keep their full rating. If S is at most G the existing connection is sufficient and the project is classified as Type 1. Otherwise, the uncovered load $S - G$ is compared against the local LV connection threshold: if it fits within LV limits and the nearest MV/LV transformer has spare capacity for it, a simple LV upgrade suffices (Type 2), and if not a new MV connection is needed (Types 3 and 4). Once the type is known, only the relevant cost components are activated, and fields that do not apply are grayed out in the interface automatically.

Project type classification. The project type depends on whether the existing grid connection can supply the planned load. To define it, the tool compares the total planned load (simultaneity × chargers × power / power factor) against the existing grid capacity and assigns one of four project types. In Type 1, the connection already has enough capacity, so only the chargers need to be installed. Type 2 corresponds to a small LV upgrade or a new LV connection that covers the gap. Type 3 indicates that a transformer exists on site but cannot handle the load, so a new MV connection is needed alongside it. Type 4 represents a site that has only a basic LV connection and needs a full MV buildout from scratch. The type determines which cost components are included and which are zeroed out. Figure 2 shows the decision logic.

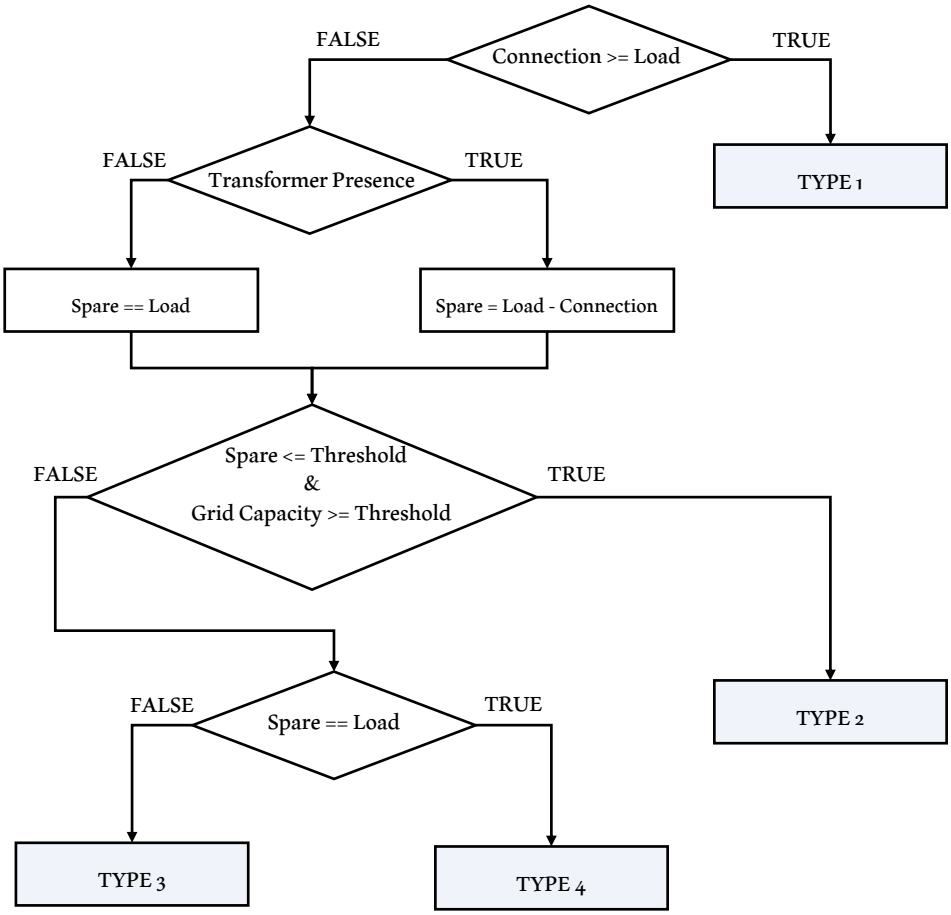


Figure 2: Flowchart for automatic project type classification.

Electrical system sizing and investment cost estimation. The tool first sizes the electrical system from the grid connection point to the charging plug (Figure 3), then estimates the investment cost as 15 items in three groups. Figure 4 shows three representative examples of how inputs flow through the pipeline to produce a cost in euros, ranging from a simple passthrough (Chargers) to a linear sizing calculation (Transformer) to a chained function (MV Connection Cost, which takes the MV cable material cost as its own input).

The key intermediate calculations that size electrical components are shown in Table 2, and the cost functions that map those quantities to euros are listed in Table 3.

Quantity	Formula
Planned apparent load	$S = f \cdot n \cdot P/\ \text{pf}$ [kVA]
LV current	$I = \frac{n \cdot P \cdot 1000}{\sqrt{3} \cdot V_{\text{LV}}}$ [A]
MV current (Type 3)	$I = \frac{S-G}{\sqrt{3} \cdot V_{\text{MV}}}$ [A]
MV current (Type 4)	$I = \frac{S}{\sqrt{3} \cdot V_{\text{MV}}}$ [A]
Transformer rating	$S_T = (S - G) \cdot (1 + \text{margin}\ /100)$ [kVA]
Rectifier modules	$k = \lceil n \cdot P/200 \rceil$

Table 2: Key intermediate calculations for electrical component sizing.

Component	Formula type	Key formula
LV/DC cables	Linear	$C = 0.276 \cdot I \cdot \text{dist}$ [€]
MV cables	Exponential	$C = 53.976 \cdot e^{0.0034 \cdot I} \cdot \text{dist}$ [€]
LV cabinet	Linear	$C = 10.319 \cdot I - 900.26$ [€]
Rectifier	Step	$C = k \cdot 44,400$ [€]
Transformer	Lookup table	S_T [kVA] $\rightarrow$ catalog price [€]
MV switchgear	Weighted linear	$C = \text{base} + \sum w_i x_i$ [€]
Surge arresters	Lookup table	V_{MV} [kV] $\rightarrow$ catalog price [€]
Charger planning	Linear	$C = n \cdot (25.064 \cdot P + 275.07)$ [€]
Charger installation	Linear	$C = n \cdot (155.83 \cdot P + 5822.5)$ [€]
Site preparation	Lookup table	$C = A \cdot r$, $r \in \{45.2, 55.2, 150.6\}$ [€/m^2]

Table 3: Cost functions mapping electrical quantities to euros.

On the MV and substation side, the model prices MVAC cables through an exponential fit to current and length, sizes the transformer through power-law scaling from apparent power [15], rates the MV switchgear through a weighted score on voltage, current, and configuration, and prices surge arresters and grounding resistors from a voltage-indexed lookup table [16]. The LV equipment group covers LVAC distribution cables applicable to Type 2 only and priced linearly on current, the LV switchboard also priced linearly on current, the rectifier at 222 €/kW in 200 kW modules and absent for charger power below 40 kW, LVDC cables from the rectifier to the chargers, and the chargers or DC dispensers themselves priced through an interpolated lookup over the 10 to 350 kW range [4]. Labor and site work are captured through linear regressions for charger planning and installation costs based on charger count and power [4,6], cable installation costs derived from expert interview data and cable material cost [17], and site preparation at a fixed rate per square metre depending on pavement material and terrain type. Any component that does not apply to the selected project type returns zero.

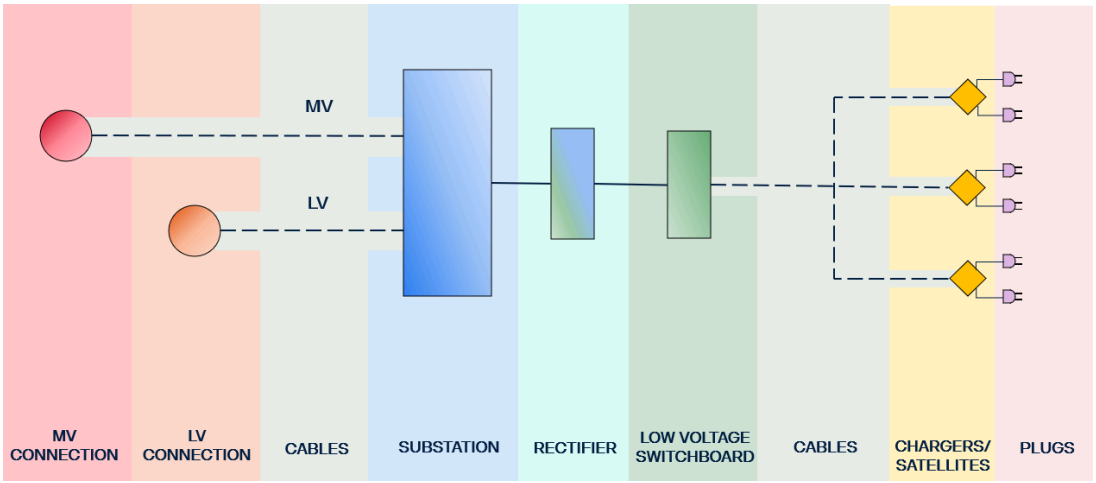


Figure 3: General layout of a high-power charging station with centralized conversion.

Analysis functions. The user can run three analysis functions: Cost Breakdown, Single-Variable Sensitivity, and Two-Parameter Sensitivity. Cost Breakdown produces a full itemized table for the exact configuration, exportable to Excel. Single-Variable Sensitivity sweeps one parameter across a user-defined range while holding everything else fixed. Because the swept parameter passes through the full pipeline on every step, increasing the number of chargers, for example, can shift the project type mid-sweep from Type 1 to Type 3 once the planned load exceeds the grid connection capacity, which in turn activates a different set of cost components; the resulting stacked bar chart makes these discontinuities visible. Two-Parameter Sensitivity sweeps two parameters at once and displays the results as a scatter plot color-coded by the second parameter, with full component breakdowns on hover. In both sensitivity analyses, every evaluation re-runs the complete pipeline covering project type classification, electrical sizing, and cost computation, so the output reflects the true non-linear response of the cost model to input variation.

Results export. The user can export every analysis to Excel and generate a PDF report with all inputs, tables, and charts included.

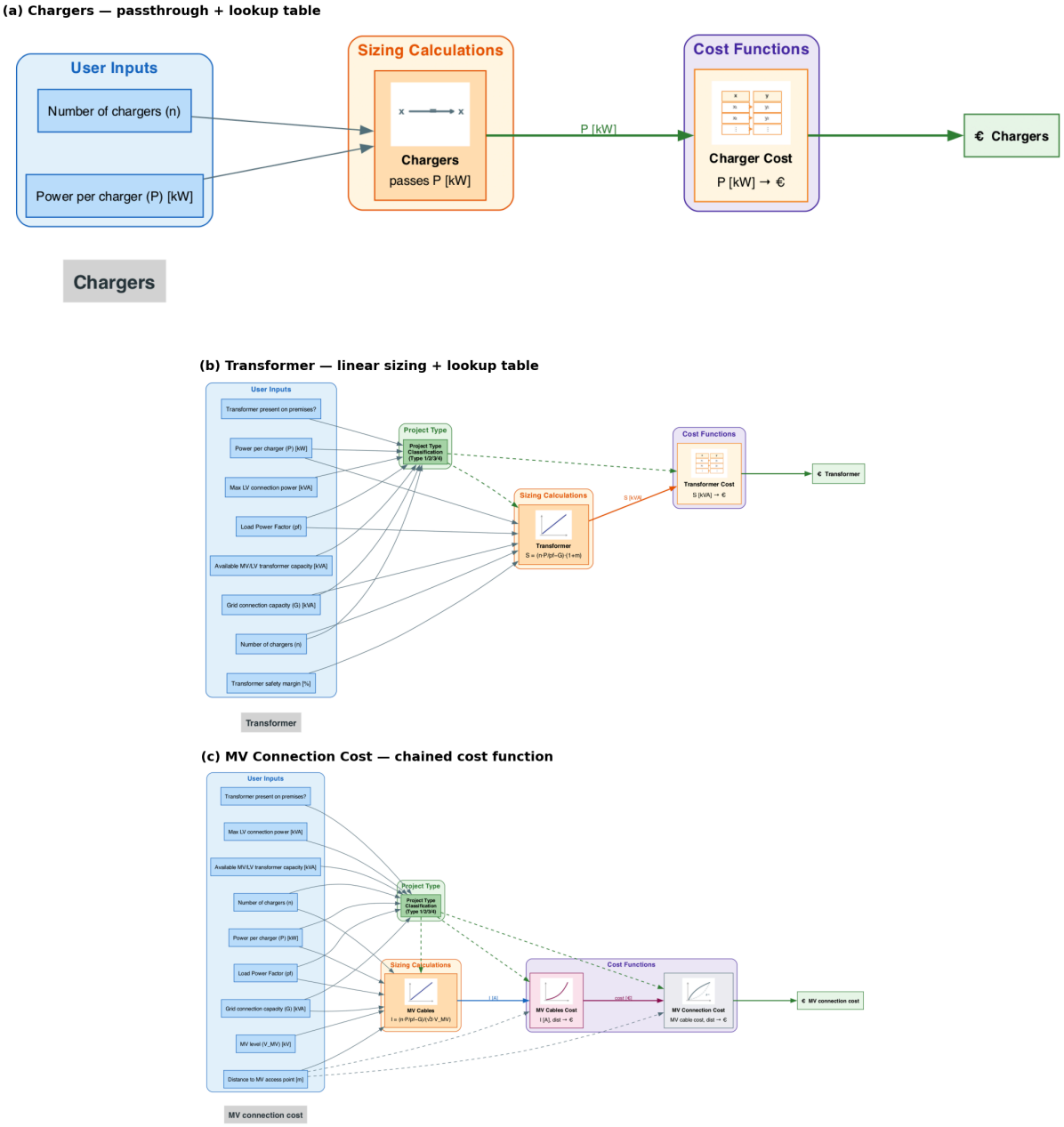


Figure 4: Three representative cost item pipelines. (a) Chargers: inputs n and P pass directly through a lookup table. (b) Transformer: six inputs drive a linear sizing calculation producing S [kVA], which indexes a catalog lookup table. (c) MV Connection Cost: a chained function that takes the MV cable material cost as input rather than raw electrical quantities.

3 Illustrative Examples

The tool is applied to nine real sites across Germany, Italy, and the United Kingdom, chosen to cover all four project types at different scales. Site selection started from a spatial analysis that cross-referenced truck stop locations [18] with nearby high-power chargers from Open Charge Map [19], specifically CCS stations of at least 100 kW within 200 meters of a truck stop. The logic is that if high-power chargers already exist nearby, there is likely MV grid capacity in the area. Table 4 lists the nine cases.

Case	Type	Country	Chargers	Power [kW]	Grid [kVA]	Transformer
1IT	1	Italy	3	200	650	Yes
3IT	3	Italy	20	160	1800	Yes
4IT	4	Italy	10	200	200	No
1DE	1	Germany	20	160	2100	Yes
2DE	2	Germany	4	100	30	No
4DE	4	Germany	40	20	50	No
2UK	2	UK	10	200	1800	Yes
3UK	3	UK	30	160	2100	Yes
4UK	4	UK	20	160	200	No

Table 4: Overview of the nine case studies.

Results span a wide range: roughly €20,000 per plug for large Type 1 sites down to over €80,000 for small Type 4 projects (Figure 5). The pattern is consistent across countries. Specific cost (€/kW) drops as installed power grows, but the savings flatten out at higher power. For Type 1 projects, chargers, rectifiers, and installation labor make up most of the bill, and specific cost falls from above 1,500 €/kW at 10 kW to below 800 €/kW at 150 kW. Type 4 projects carry the extra burden of a full MV buildout (transformer, switchgear, MV cables, trenching), which does not scale as well; their specific cost drops to about 1,000–1,100 €/kW and then levels off. The gap between Type 1 and Type 4 at the same power level is roughly 60% MV equipment and 40% additional labor.

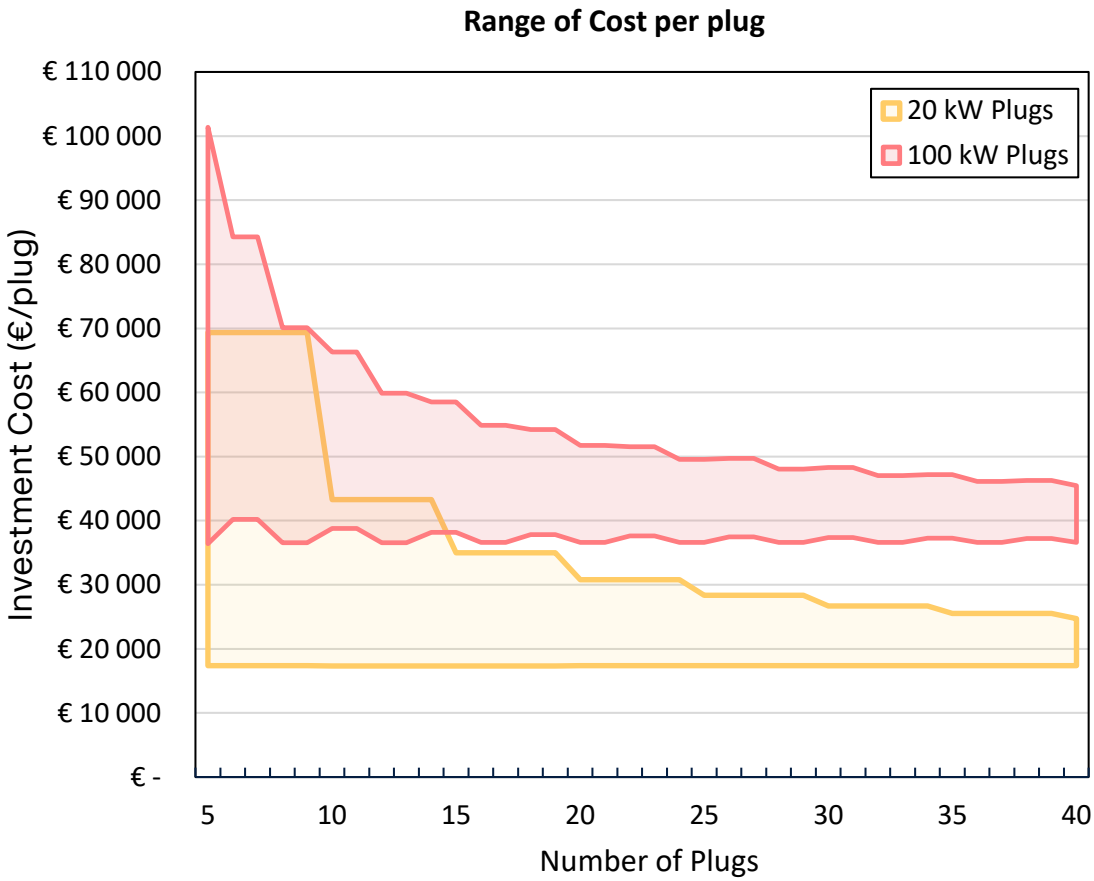


Figure 5: Investment cost per plug as a function of deployment scale, for 20 kW and 100 kW chargers.

Figure 6 breaks down each case by component. Case 3UK costs the most in absolute terms (30 chargers at 160 kW with a new MV connection), but its per-kW cost is among the lowest because the fixed MV costs are spread over 4.8 MW of capacity. Smaller projects like 2DE and 1IT stay under €500,000 total but pay more per kilowatt.

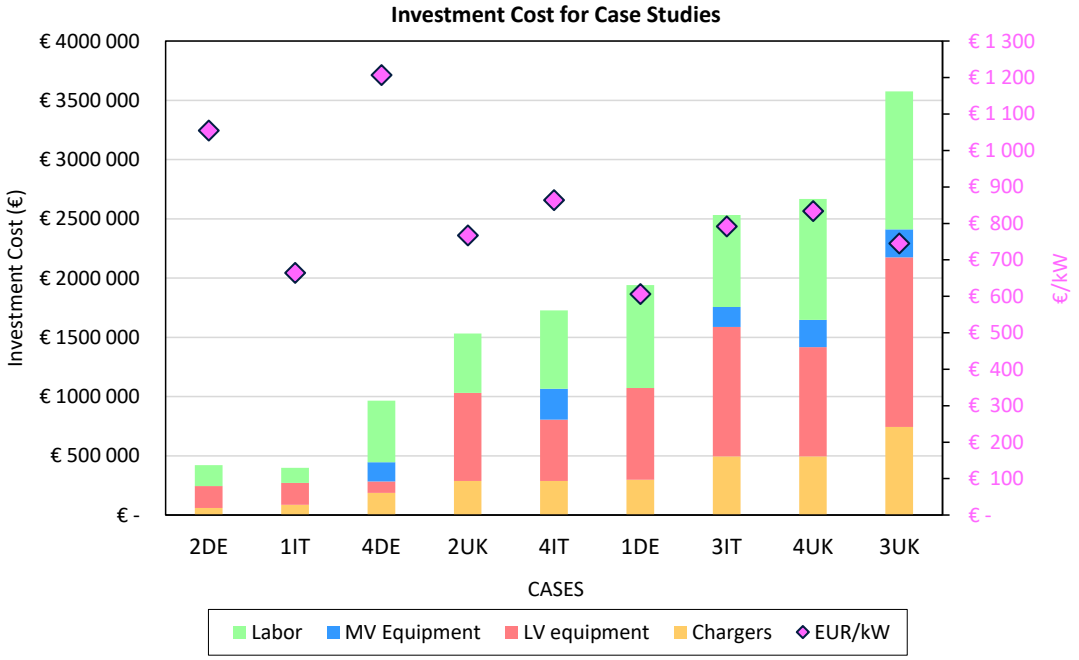


Figure 6: Investment cost and specific cost (€/kW) for the nine public parking case studies.

4 Impact

The tool provides decision-makers with a fast, transparent way to estimate and compare overnight HDV charging infrastructure costs before committing to a project. Fleet operators, infrastructure developers, and policymakers can use it to understand how site-specific design choices such as charger count and power, grid connection type, and cable distances drive total investment cost. AFIR requires over 300,000 truck charging points across Europe by 2030 at an estimated €7 billion [7,8], and the range of per-plug costs found here (€20,000–€80,000) shows that informed design choices can make a substantial difference in how that budget is spent.

In practice, an operator can compare high-power against low-power chargers, check whether an LV upgrade is sufficient or an MV connection is unavoidable, and see which parameters move the total cost most. Policymakers can draw on the same results to design subsidy schemes, set grid connection standards, or decide which TEN-T corridor locations to prioritize for public charging infrastructure.

For researchers, the value lies in the open, modular structure of the codebase. Every cost function can be inspected, modified, or replaced with local data, and if MCS connector pricing becomes available or a user wants to substitute labor rates for a specific country, the relevant lookup tables can be edited directly [20,21]. The tool was developed in collaboration with Scania AB, with cost assumptions informed by interviews with nine industry professionals [17]. On the operational side it complements FleetRL [14], so that together the two tools cover both the capital investment and the ongoing operational costs of electric truck depot charging.

5 Conclusions

This paper has introduced an open-source tool that estimates the investment cost of overnight HDV charging infrastructure in Europe, covering 15 cost components from the grid connection to the plug. It assesses each site's grid connection requirements and diagnoses which of the four project types needs to be implemented, then supports cost breakdown, single-variable, and two-parameter sensitivity analysis through a web interface.

Across nine case studies in Germany, Italy, and the UK, costs range from about €20,000 per plug (large Type 1 sites with existing grid capacity) to over €80,000 (small Type 4 sites needing a full MV buildout). Economies of scale are real but have a floor: MV equipment adds a fixed layer that does not shrink much with more chargers. For Type 1 projects, chargers and installation dominate; for Type 3 and 4, the transformer, switchgear, and MV cabling become the main differentiator.

Future work will extend the tool in several directions. The most significant addition would be electricity cost optimization and levelized cost of charging (LCOC) calculations, which would allow users to compare not just the capital investment but the full cost of ownership across configurations. Shorter-term extensions include lead time estimation by project type, updated pricing for Megawatt Charging System (MCS) connectors as that standard matures in the European market, and tools for scaling the methodology from individual sites to regional deployment planning.

CRediT Authorship Contribution Statement

Davide Ferraro: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Visualization, Validation. **Maria Carolina Gil Ribeiro:** Writing – review & editing. **Jagruti Thakur:** Writing – review & editing, Supervision, Methodology, Conceptualization, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The source code is available at https://github.com/davide-ferraro/ev-charging-cost-estimator. The full methodology and data sources are documented in the accompanying master's thesis report, available at https://kth.diva-portal.org/smash/record.jsf?pid=diva2%3A2007635.

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